La Scuola d'Italia / Dr. Huson / IB Math 12
21 October 2026

Nickname:

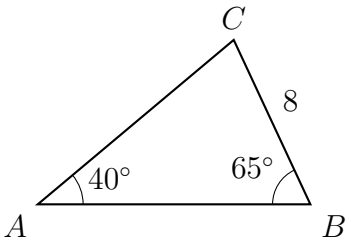
2.3 Classwork: The Sine Rule and the Area of a Triangle

In triangle ABC, side a is opposite angle A, side b is opposite angle B, and so on.

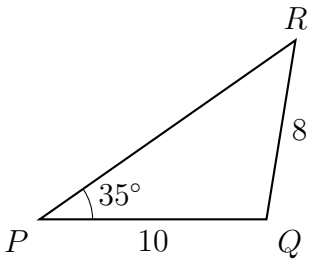
$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} \qquad \text{Area} = \frac{1}{2}ab\sin C$$

Give answers correct to three significant figures, unless they are exact.

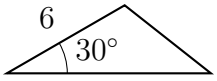
1. The diagrams are not to scale.



(a), (b)



(c)



(d)

(a) Find the length AC.

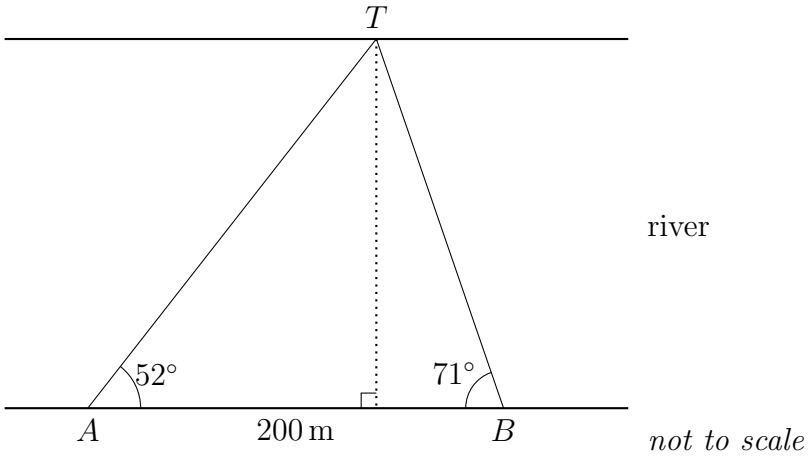
(b) Find the area of triangle ABC.

(c) In triangle PQR, angle R is acute. Find the size of angle R.

(d) Find the exact area of the third triangle.

(e) A different triangle has two sides of length 5 cm and 10 cm, and its area is 20 cm^2. The angle between these two sides is acute. Find the size of this angle.

2. Points A and B are 200 m apart on the straight bank of a river. A tree T stands on the opposite bank. Angle $TAB = 52°$ and angle $TBA = 71°$. The two banks are parallel.



(a) Write down the size of angle ATB.

(b) Find the distance AT.

(c) Find the width of the river.

(d) Find the area of triangle ABT.

A triangular garden has two sides of length 8 m and 10 m. Its area is 30 m^2, and the angle between these two sides is acute.

(e) Find the size of the angle between the two sides.